Priyansh Gupta

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SUMMARY

Graduate **Machine Learning Engineer** with strong foundations in **algorithms**, **systems**, and **end to end ML development.** Skilled in **Python**, **C++,** and **applied ML.** Experienced in optimization, model design, data pipelines, and analytical reasoning. Motivated to work at the intersection of ML software, compilers, and hardware aware systems.

Key Achievements

- Published a **Peer-reviewed research** on Gaussian mixture model clustering at LINHAC 2025 (DOI).
- Finalist of Student Research Competition at LINHAC 2024(Proceedings, p.100)
- Received Cisco security Brown belt

WORK EXPERIENCE

Research Assistant
Linköping University, Sweden | Oct 2025 – Dec 2025

- Mentored junior researchers on Machine Learning methodologies, specifically guiding the application of probabilistic modeling to high-frequency sensor data in Handball and Ice Hockey.
- Orchestrated a Journal-track submission by extending thesis research with advanced player filtering to identify playing style patterns.

Master's Thesis - Data Science & ML
Linköping University, Sweden | Jan 2025 – Sep 2025

- Designed scalable ML + data pipelines (Python, Dask, SQLite) processing **29M+ time-series records**, reducing preprocessing time from **5h → 40 min**.
- Applied unsupervised probabilistic modeling (**Gaussian Mixture Models)** for behavioural segmentation, uncovering latent movement and interaction patterns relevant for abstract reasoning.
- Built an automated model-selection engine (15K+ configs) using BIC, increasing reproducibility and reducing tuning time by 80%.
- Delivered technical results at LINHAC 2025.

Research and Teaching Assistant - Data Science
Linköping University, Sweden | Oct 2024 – Dec 2024

- Developed predictive ML models, improving AUC by 24% using weighted loss optimization.
- Built statistical validation and SHAP explainability workflows.
- Managed full analytics lifecycle: experimentation, validation, and reporting.

Software Engineer
Cisco Systems, India | Jan 2020 – Jul 2023

- Built telemetry-driven analytics pipelines enabling proactive anomaly detection, improving reliability by 40%.
- Developed microservices (**Java**, **PostgreSQL**) with CI/CD automation, improving deployment robustness by 75%.
- Collaborated with cross functional engineering teams using **agile** and **version control**.

RELEVANT PROJECTS

Sokoban Puzzle Solver (AlphaZero-inspired RL) | GitHub — **RL, CNN, MCTS**

Developed an AlphaZero-style RL agent using **MCTS + CNN** value network for puzzle solving, with **cycle/softlock detection** and heuristic learning to improve search efficiency.
Demonstrated strong **algorithmic reasoning** and **systems level** understanding.

Property Price Prediction — **AWS, Random Forest, XGBoost**

Engineered spatial & structural features and developed robust real-estate pricing models with strong cross-city generalization.

TECHNICAL SKILLS

ML & Statistics: Predictive modeling, Unsupervised learning, Time-Series analysis, Scikit-learn, PyTorch, TensorFlow
Data Eng. : Pandas, Spark, Airflow, ETL, PostgreSQL
Programming: Python, SQL, R, Java, C++

EDUCATION

Linköping University, Sweden | **M.Sc. in Computer Science – AI and Data mining | GPA:** 4.3/5.0
SRM Institute of Science and Technology, India | **B. Tech in Computer Science | CGPA:** 8.9/10

PUBLICATIONS AND AWARDS

- **Peer-reviewed research** on GMM clustering, published at LINHAC 2025 (DOI).
- **Finalist**, LINHAC 2024 Student Research Competition (Proceedings, p.100)
- Microsoft Platinum Badge (**Agentic AI, LLMs**)

POSITIONS OF RESPONSIBILITY

- **Organising Committee Member,** Linköping Hockey Analytics Conference 2025 (LINHAC25)
- Treasurer and Head of Sponsorship, **International Student Association** (Linköping University)